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LondheShubham153 added day 21

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Day 21 – Shell Scripting Cheat Sheet: Build Your Own Reference Guide

Task

You've spent the last several days learning Shell scripting — from basics to real-world projects. Now it's time to consolidate everything into a **personal cheat sheet** that you can use as a quick-reference guide for the rest of your DevOps journey.

The best way to revise is to **teach it back**. Writing a cheat sheet forces you to organize your understanding and identify gaps.

Expected Output

- A markdown file: `shell_scripting_cheatsheet.md`

Challenge Tasks

Task 1: Basics

Document the following with short descriptions and examples:

1. Shebang (`#!/bin/bash`) — what it does and why it matters
2. Running a script — `chmod +x` , `./script.sh` , `bash script.sh`
3. Comments — single line (`#`) and inline
4. Variables — declaring, using, and quoting (`$VAR` , `"$VAR"` , `'$VAR'`)
5. Reading user input — `read`
6. Command-line arguments — `$0` , `$1` , `$#` , `$@` , `$?`

Task 2: Operators and Conditionals

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1. String comparisons — `=` , `!=` , `-z` , `-n`
2. Integer comparisons — `-eq` , `-ne` , `-lt` , `-gt` , `-le` , `-ge`
3. File test operators — `-f` , `-d` , `-e` , `-r` , `-w` , `-x` , `-s`
4. `if` , `elif` , `else` syntax
5. Logical operators — `&&` , `||` , `!`
6. Case statements — `case ... esac`

Task 3: Loops

Document with examples:

1. `for` loop — list-based and C-style
2. `while` loop
3. `until` loop
4. Loop control — `break` , `continue`
5. Looping over files — `for file in *.log`
6. Looping over command output — `while read line`

Task 4: Functions

Document with examples:

1. Defining a function — `function_name() { ... }`
2. Calling a function
3. Passing arguments to functions — `$1` , `$2` inside functions
4. Return values — `return` vs `echo`
5. Local variables — `local`

Task 5: Text Processing Commands

Document the most useful flags/patterns for each:

1. `grep` — search patterns, `-i` , `-r` , `-c` , `-n` , `-v` , `-E`
2. `awk` — print columns, field separator, patterns, `BEGIN/END`
3. `sed` — substitution, delete lines, in-place edit
4. `cut` — extract columns by delimiter
5. `sort` — alphabetical, numerical, reverse, unique
6. `uniq` — deduplicate, count
7. `tr` — translate/delete characters
8. `wc` — line/word/char count
9. `head` / `tail` — first/last N lines, follow mode

Task 6: Useful Patterns and One-Liners

Include at least 5 real-world one-liners you find useful. Examples:

- Find and delete files older than N days
- Count lines in all `.log` files
- Replace a string across multiple files
- Check if a service is running
- Monitor disk usage with alerts
- Parse CSV or JSON from command line
- Tail a log and filter for errors in real time

Task 7: Error Handling and Debugging

Document with examples:

- 1. Exit codes — `$?` , `exit 0` , `exit 1`
- 2. `set -e` — exit on error
- 3. `set -u` — treat unset variables as error
- 4. `set -o pipefail` — catch errors in pipes
- 5. `set -x` — debug mode (trace execution)
- 6. Trap — `trap 'cleanup' EXIT`

Task 8: Bonus — Quick Reference Table

Create a summary table like this at the top of your cheat sheet:

Topic	Key Syntax	Example
Variable	<code>VAR="value"</code>	<code>NAME="DevOps"</code>
Argument	<code>\$1</code> , <code>\$2</code>	<code>./script.sh arg1</code>
If	<code>if [condition]; then</code>	<code>if [-f file]; then</code>
For loop	<code>for i in list; do</code>	<code>for i in 1 2 3; do</code>
Function	<code>name() { ... }</code>	<code>greet() { echo "Hi"; }</code>
Grep	<code>grep pattern file</code>	<code>grep -i "error" log.txt</code>
Awk	<code>awk '{print \$1}' file</code>	<code>awk -F: '{print \$1}' /etc/passwd</code>
Sed	<code>sed 's/old/new/g' file</code>	<code>sed -i 's/foo/bar/g' config.txt</code>

Format Guidelines

Your cheat sheet should be:

- Written in **Markdown** (`.md`)
- Organized with **clear headings** for each section
- Include **code blocks** with syntax highlighting (````bash`)
- Keep explanations **short** — 1-2 lines max per item
- Focus on **practical examples** over theory
- Something **you would actually refer back to** on the job

Submission

1. Add your `shell_scripting_cheatsheet.md` to `2026/day-21/`
2. Commit and push to your fork

Learn in Public

Share your cheat sheet on LinkedIn — help others revise too!

#90DaysOfDevOps #DevOpsKaJosh #TrainWithShubham

Happy Learning! **TrainWithShubham**